F03 Variation, proportions and probability

Week 3, fourth teaching day. Complete ML09 first.

Today's question. Two groups can have the same mean. How can I tell them apart?

WHERE THIS IS USED

Forecasts speak in chances, and clinics report the share of patients with a problem. Two groups can have the same average and still look very different. How would you tell two football teams apart if both score the same average?

Words for today

Word	What it means	Think of it like this
mean	The total shared out equally.	Sharing a bag of mandazi.
deviation	A value minus the mean; it can be negative.	How far a mark is from the class average.
standard deviation	A typical size of the deviations, in the original unit.	How spread out the scores are.
proportion	A part divided by the whole.	3 blue counters out of 5.
probability	How likely an outcome is under stated conditions.	A fair die gives a 6 about once in six throws.
observed fraction	What actually happened in a limited number of tries.	Ten throws of a die.

Learn the idea

The mean is a total divided by a count. Two groups can have the same mean and different spread. A deviation is a value minus the mean. NumPy finds the mean and the standard deviation of an array: values.mean() and values.std(). Here the numbers describe the whole small group. A proportion is a part divided by a whole, and times 100 it becomes a percentage. A probability describes a chance under stated conditions. An observed fraction estimates it and does not promise the next outcome.

Start on paper

Put four cards on the table: 2, 2, 6, 6. The mean is 4. The deviations are -2, -2, 2, 2. The standard deviation is 2. Now four cards 4, 4, 4, 4: the same mean, no spread. For a fair six-sided die two of six faces are above 4, so the probability is 2/6. Ten real throws need not give exactly that fraction.

Worked example

```
import numpy as np
values = np.array([2, 2, 6, 6])
mean = values.mean()
deviations = values - mean
spread = values.std()
steady = np.array([4, 4, 4, 4])
print(mean, deviations, spread)
print(steady.mean(), steady.std())
print(2 / 6, 2 / 6 * 100)
```

Run it, then compare: Line one: mean 4.0, deviations minus 2, minus 2, 2 and 2, spread 2.0. Line two, the steady group: mean 4.0, spread 0.0. Line three: 0.3333 and 33.33, a chance of 1 in 3, or 33.3 percent.

Your turn

Repeat for [1, 1, 5, 5]. Explain the mean and spread in words. A bag has 3 blue and 2 red counters. State the chance of blue and its percentage, assuming each counter is equally likely. Save F03.ipynb.

Run all cells from the top, save, close and reopen. Show the instructor your output and explanation.